ZeRO: Memory Optimizations Toward Training Trillion Parameter Models

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Abstract
This paper investigates optimization with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction
Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding ZeRO: Memory Optimizations Toward Training Trillion Parameter Models.

Section topic: Optimization
Question context: How does DeepSpeed ZeRO (Zero Redundancy Optimizer) partition training states across data-parallel processes?
Key Finding: ZeRO eliminates memory redundancy across data-parallel ranks by partitioning optimizer states (Stage 1), gradients (Stage 2), and model parameters (Stage 3) [ZeRO, p. 3].
Evidence passage: ZeRO-Stage 1: Partitions optimizer states ZeRO-Stage 2: Partitions gradients; ZeRO-Stage 3: Partitions model parameters across GPUs

2. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for ZeRO: Memory Optimizations Toward Training Trillion Parameter Models. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

3. Technical Methodology & Empirical Findings

Section topic: Multi-Paper Analysis
Question context: How does ZeRO Stage 3 parameter partitioning compare to Megatron-LM tensor model parallelism?
Key Finding: Megatron splits weight tensors across GPUs requiring high-bandwidth all-reduce steps, while ZeRO-3 broadcasts full parameters on-demand and discards them after execution [ZeRO, p. 4].
Evidence passage: Megatron splits weight matrices across GPUs within each layer, requiring frequent all-reduce communication ZeRO-3 gathers parameters dynamically per forward/backward pass and discards them immediately

4. Technical Methodology & Empirical Findings

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